University of Rajshahi
Dept. of Computer Science and Engineering
B.Sc. Engg. Part-I, Even Semester, Examination 2020
Course Code: CSE 1211 Course Title :Introduction to Digital Electronics
Time: 3 Hours **Full Marks: 52.5**
[N.B. Answer SIX questions taking at least THREE from each Section.]

Section A

1. (a) What are the advantages of encoding a decimal number in BCD as compared to straight binary? What is its disadvantage? 2
   (b) What range of decimal values can be represented by a four-digit octal number? 1.75
   (c) A typical PC uses a 20-bit address code for its memory locations- 3
      i) How many Hex digits are needed to represent a memory address?
      ii) What is the range of addresses?
      iii) What is the total number of memory locations?
   (d) Perform the subtractions $(01001 - 11010)_2$ and $(10010 - 10011)_2$ using 2's complement system. 2

2. (a) Draw the logic diagram and truth table of OR, NOR and XOR gate using NAND gate only. 3
   (b) Simplify the following Boolean expression 3
      i) $X = ABC + A'BC + AC + AC'$
      ii) $A'B\,(D' + CD) + AB + A'BCD$
   (c) Which coding technique is good for error detection? Give example. Convert $(10110)_2 = (?)_{gray}$. 2.75

3. (a) Minimize the following function: $f(a,b,c,d) = \sum a(1,3,4,7,11) + d(5,12,13,14,15)$ 3
   (b) Define prime implicant with example. 1.75
   (c) Minimize the following function using Quine-McCluskey method. 4

$$f(a,b,c,d) = \sum_{m}(0,1,2,4,6,8,12,14)$$

4. (a) Define the following terms: i) Fan out ii) Noise margin 2.75
   (b) Draw and explain the circuit operation of 2 digit TTL NAND gate. 3
   (c) Design a two input CMOS NAND gate with necessary diagram and truth table. 3

Section B

5. (a) Write down some benefits of clocked flip-flop. Discuss the circuit diagram of the edge triggered SR flip-flop with timing diagram. 4
   (b) Mention some applications of Latch. 1.75
   (c) Briefly explain the operation of master-slave flip-flop using logic and timing diagram. 3

6. (a) Draw the internal functional diagram of a 555 timer and explain its basic operation. What are the applications of 555 timer? 4.75
   (b) Draw the pin configuration of a 555 timer IC and explain the function of each pin. 4

7. (a) Define resolution or step size of a D/A converter. For a DAC, if step size is 0.1 V, then what will be the output voltage for a digital input 0001. 2.75
   (b) What is the advantage of R/2R ladder DAC over weighted registers DAC? Draw and explain the basic operation of R/2R ladder DAC. 3
   (c) What is the main advantage of a SAC over digital ramp ADC? An 8-bit SAC has resolution of 20 mV. What will its digital output be for an analog input of 2.17 V? Illustrate with diagram. 3

8. Write short notes on the following (any three): 8.75
   (a) ECL NOR/OR gate
   (b) Shift register
   (c) Digital Ramp ADC
   (d) 3 bit counter

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg.(CSE) 1st Year Even Semester 2019
Course: CSE1211 (Introduction to Digital Electronics)
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Section A

1. (a)	Perform subtraction using 2's complement method $(85\text{-}47)_{10}$.	3
(b)	convert $(541.203)_6$ to base 5, base 8 and base 10.	3
(c)	Write the binary code for the gray code 10110111.	0.75
(d)	Represent the decimal number 28 to excess-3 and BCD code.	2
2. (a)	Given the boolean function K = AB+A'B'+B'C Then i) implement it with AND, OR and NOT gates ii) implement it with only OR and NOT gates.	3
(b)	Show that the dual of the exclusive-OR is equal to its complement. Simplify the following expression $z = \bar{A}B\bar{C} + AB\bar{C} + B\bar{C}D$	3.75
(c)	What are don't care terms? Explain with example.	2
3. (a)	What is parity bit? Design 3-bit odd parity generator and checker.	6
(b)	Implement a full adder using two half adder.	2.75
4. (a)	Design a logic circuit whose output is HIGH only when a majority of inputs A, B and C are LOW.	2.75
(b)	Simplify the following Boolean function using K-map and realize with basic gates: $F(A, B, C, D) = \sum m(1, 5, 6, 12, 13, 14) + \sum d(2, 4)$.	6

Section B

5(a)	Show the logic diagram of a clocked RS flipflop with four NAND gates.	3
(b)	With the help of timing diagram, explain the operation of positive edge triggered JK FF.	3
c)	What are the advantages of a master-slave JK FF? Mention some limitations of SR FF.	2.75
6(a)	What is the basic difference between latch and flip-flop?	1
(b)	Define setup time, propagation delay and hold time for flip-flop using proper timing diagram.	2.75
(c)	Draw the circuit diagrams and discuss the operation of D flip-flop and T flip-flop.	5
7.(a)	Draw the TTL NAND gate circuit and explain its operation.	3
(b)	What is current sourcing and current sinking action?	1.75
(c)	Draw and explain a CMOS NOR gate circuit.	4
8.(a)	Discuss the operation of a 555 timer IC based monostable multi-vibrator.	3
(b)	A certain binary-weighted-input DAC has a binary input of 1101. If a HIGH = +3.0 V and a LOW = 0 V, what is V_{out}?	1.75
(c)	Draw the block diagram of a successive-approximation ADC and discuss its basic operation with a flow-chart.	4

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[N.B. Answer SIX questions taking at least THREE from each Section.]

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Section A

1. (a) Find $(45)_{10} - (83)_{10}$ using two's complement format with 8-bit numbers. Then convert your result back to decimal. 3
   (b) Add $(65)_{10} + (72)_{10}$ using 8-bit sign-magnitude format for the numbers. Convert your result to decimal. Is your answer correct? Why or why not? 5
   (c) Write the gray code for $(1011)_2$. 0.75

2. (a) Convert $(1000\ 0110)_{BCD}$ to decimal, binary & octal. 3
   (b) Simplify Z = A' C (A' B D)' + A' B C' D' + A B' C using Boolean algebra. 2.75
   (c) State & prove De Morgan's theorems with the help of truth tables. 3

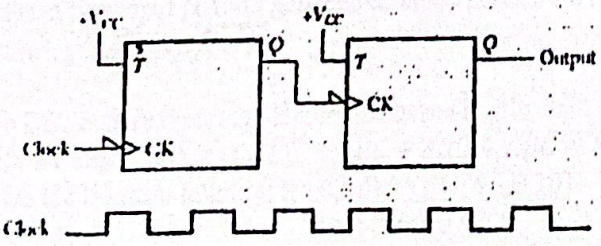
3. (a) Design and explain a full adder in detail with circuit diagram and truth table. 4
   (b) Design a combinational logic circuit to compare two 2-bit binary numbers A and B and to generate the outputs A<B, A=B and A>B. Is there a way to derive the third output from the first two outputs? 4.75

4. (a) Simplify the following Boolean expression using Quine-McCluskey technique 6
   f(A,B,C,D)= Σm(0,1,3,7,8,9,11,15).
   (b) List out the advantages and disadvantages of Quine-Mc Cluskey Method. 2.75

Section B

5(a) What is latch and flip-flop? 2
   (b) Explain the master slave S-R flip-flop with timing diagram in detail. 6.75

6(a) Draw a circuit diagram of a J-K latch and explain its operation. 4
   (b) Plot the output waveforms referenced to the clock signal assuming the initial contents of all FFs is Q = 0. Assume all FFs are edge triggered. 4.75



7.(a) State advantages and disadvantages of TTL. 2
   (b) Draw the circuit diagram of DTL-NAND gate. 3
   (c) Design a NOT gate using MOSFET. 3.75

8.(a) Draw the block diagram of a system to interface a computer to the analog world so that the computer can monitor and control a physical variable and explain the functions of each block. 4
   (b) What do you mean by resolution and accuracy for a DAC? 1.75
   (c) Draw the block diagram of an ADC and discuss its basic operation. 3

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg. Part-I Even Semester 2015
Course: CSE1211 (Introduction to Digital Electronics)
Time: 3 Hrs. Full Marks: 52.5
[N.B. Answer SIX questions taking at least THREE from each Section.]

Part A

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1. (a) Do the following conversions: 4
   i) $(378)_{10}$ to 16 bit binary number;
   ii) $(1010110.1100)_2 = (?)_{10}$;
   iii) $(10101100)_2 = (?)_8$;
   iv) $(743)_{16} = (?)_2$.
   (b) Compare between BCD and Binary code. 2
   (c) What is Gray code? Explain with examples, how do you convert binary to Gray and Gray to binary? 2.75

2. (a) Subtract $(100111)_2$ from $(001100)_2$ using 2's complement method. Why do you need 2's complement method? 2
   (b) Apply the input waveforms of fig-1 to a NOR gate, and draw the output waveform. Then repeat the output waveform with C hold permanently LOW. 2

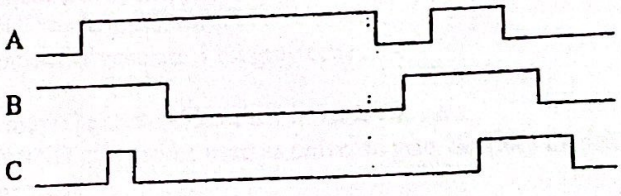


Fig-1

(c) Determine the truth table for the circuit of fig-2. 2

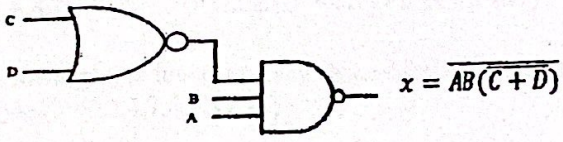


Fig-2

(d) Show that a two-input NAND gate can be constructed from two-input NOR gate. Simplify the following: $(\overline{A+B})(\overline{\bar{A}+\bar{B}})$ 2.75

3. (a) What is DeMorgan's theorem? Explain with truth table. 2.75
   (b) Simplify the expression $x = \bar{A}\bar{B}\bar{C} + \bar{A}BC + ABC + A\bar{B}\bar{C} + A\bar{B}C$ using Boolean algebra. 2
   (c) Minimize the Boolean function $f(a,b,c,d) = \sum_M(1,4,6,8,10)$ 4

4. (a) Simplify the following expression using K-map method 5
   $f(A,B,C,D) = \sum_M(7, 9, 10, 11, 12, 13, 14, 15)$
   (b) Design the circuit corresponding to the truth table shown in Table-1. 3.75

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Table-1

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Part B

5. (a)	Explain the clocked S-R flip-flop in detail.	6
(b)	Explain D type flip-flop.	2.75
6. (a)	Draw the basic DTL gates to implement NAND gate. Explain its operation.	2
(b)	Discuss the characteristic of TTL gates. Explain the operation of open collector TTL gates.	2.75
(c)	Where ECL is more suitable? Draw and explain ECL to implement OR and NOR gates.	4
7. (a)	Define transducer, analog-to-digital converter (ADC) and digital-to-analog converter (DAC).	3
(b)	Draw the basic R/2R ladder DAC of 4-bit and write the V_{OUT} expression.	2
(c)	Assume the V_{REF} =5V for the 4-bit DAC. What are the resolution and full-scale output of this converter?	1.75
(d)	An 8-bit DAC has an output of 3.92 mA for an input of 01100010. What are the DAC's resolution and full-scale output?	2
8. (a)	What is timer? What are the different applications of timer?	3
(b)	Design an monostable multi-vibrator using 555 timer and explain its operation.	5.75

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Department of Computer Science and Engineering
B.Sc. Engg. (CSE) Part-1 Even Semester Exam-2014
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Part-A

1. (a) Do the following conversions: 4
   (i) $(10101011.1101)_2=(?)_{10}$ (ii) $(20345.125)_{10}=(?)_2$ (iii) $(80914.25)_{10}=(?)_8$
   (iv) $(3AE8F.2D)_{16}=(?)_8$ (v) $(1011001110)_2=(?)_4$
   (b) Add $(110111)_2$ with $(100111)_2$. Subtract $(100110)_2$ from $(110011)_2$ using 2's complement method. 3
   (c) Represent $(-17)_{10}$ in sign magnitude, 1's complement and 2's complement representation. 1.75

2. (a) Write the procedure to convert a binary code to gray code with example. 4
   (b) With the help of example explain excess-3 code. 3
   (c) Write the BCD code for $(9248)_{10}$. 0.75
   (d) How can you easily generate 3 bit gray code . 1

3. (a) Define and draw the truth table of a 3-input X-OR gate. 2.75
   (b) Show that NAND gate can be used as universal gate. Simplify the following using De-Morgan's theorem 3
   $\overline{(A+\overline{B})}\,\overline{(\overline{A}+B)}$
   (c) Simplify the following Boolean expressions to a minimum number of literals: 3
   (i) $ABC+A'B+ABC'+AC$ (ii) $A'B(D'+\overline{CD})+B(A+A'CD)$

4. Simplify the following logic function using Quine-McCluskey technique 8.75
   $f(A,B,C,D) = \sum m(0,1,3,7,8,9,11,15)$

Part-B

5. (a) Write the difference between asynchronous and synchronous system. 1.75
   (b) Explain the operation of a positive-edge triggered SR flip-flop by timing diagram. 4
   (c) Discuss how an SR latch is converted into D-latch. 3

6. (a) Explain the operation of a JK flip-flop using timing diagram. 3.75
   (b) Mention the limitations of a JK flip-flop. 2
   (c) Describe how a D-latch operates differently from an edge-triggered D flip-flop. 3

7. (a) Discuss the operation of an 8-bit DAC using op-am summing amplifier with binary weighted registers. 3
   (b) Define R/2R ladder DAC with basic circuit diagram. Write the benefits of R/2R circuit. 3
   (c) Briefly discuss the operation of the digital ramp ADC. 2.75

8. (a) Draw the block diagram of a 555 timer and explain about its each pin. 3
   (b) Design an astable multi-vibrator using 555 timer and explain its operation. 5.75

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